

Open-Source Software Indicators for the Science and Engineering Indicators 2026

Technical Documentation

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1. Introduction

Science and engineering activity is increasingly embedded in and documented through software tools that are freely available to use, reuse, and modify. Open source software (OSS) allows free access to modifiable digital tools that are used for work and leisure and constitutes a part of intangible investment with the qualities of knowledge-based public goods. Indicators of its creation and use illustrate aspects of innovation and technology diffusion that enhance science and technology indicators. The National Science Board's Science and Engineering Indicators (SEI) report first introduced experimental measures of OSS activity for government, academic institutions, and businesses, and for countries where contributors reside in 2022 (NSB 2022).

Westat has been commissioned by the National Center for Science and Engineering Statistics (NCSES), on behalf of the National Science Foundation (NSF), to develop measures and indicators of OSS activity for inclusion in the *Indicators 2026* report "Translation to Impact: U.S. and Global Science, Technology, and Innovation Output" (NSB 2026a). This technical document describes the data sources, processing pipeline, and methodological steps used to construct the OSS indicators, including data collection, cleaning, standardization, and statistical production.

Westat created the OSS indicators using data collected from GitHub, a platform that allows software developers to create, store, modify, and share software. The GitHub data are used in this report for three reasons:

- GitHub's focus on modification and sharing provides the opportunity to collect metadata about software repositories and its creators.
- GitHub is currently very broadly used with over 420 million repositories (GitHub 2026a).
- Alternative sources of data for federal contributions, such as code.gov, have been discontinued. 3—4

The *Indicators 2026* report provides indicators on GitHub repositories as the unit of observation (NSB 2026a). A software developer can contribute code to a project on GitHub by performing a "git commit" (GitHub 2026b), and these contributions to an open-source licensed project in a repository in a given year are a measure of innovation activity. For this analysis, OSS is computer software, with its source code made available with a license, in which the copyright holder provides the rights to study, change, and distribute the software to anyone and for any purpose and which is identified by the Open Source Initiative (OSI) (<https://opensource.org/licenses>). When available, data identifying the country or economic sector of the developer making the contribution are an indicator of the source of this innovation activity.

The OSS indicators in the report quantify (i) GitHub repositories contributed to in a given year, (ii) geographical location of repository contributors, (iii) institutional affiliation of repository contributors, and (iv) collaborations between contributors from different country pairs. The figures associated with these indicators are provided in the *Indicators 2026* report's sidebar "Global Contributions and Collaborations in Open-Source Software." Table 1 below illustrates

the figures and the respective supplemental tables provided in Translation to Impact: U.S. and Global Science, Technology, and Innovation Output: Supplemental Tables (NSB 2026b).

Table 1. Figures and Supplemental Tables	
Figures	Supplemental Tables
Figure TRN-A. New GitHub repositories contributed by the United States and the rest of the world: 2009–24	Table STRN-33. Contributions to New GitHub repositories by selected entities: 2009–24
Figure TRN-B. Share of new GitHub repositories, by region, country, or economy of contributor: 2024	
Figure TRN-C. Open-source software collaboration networks in GitHub, by region, country, or economy: 2024	Table STRN-35. Open-source software collaboration networks in GitHub, by region, country, or economy: 2024
	Table STRN-36. Open-source software repositories contributed to in GitHub, by country of contributor: 2024

This technical appendix is organized as follows. Section 2 describes the data and methods. Section 3 describes the supplemental tables included in the *Indicators 2026* report “Translation to Impact: U.S. and Global Science, Technology, and Innovation Output” (NSB 2026a). Section 4 describes the methodological steps used to construct the figures and supplemental tables, including data collection, cleaning, standardization, and statistical production. Section 5 discusses limitations and concludes.

2. Data, Methods, and Limitations

While the sharing of computer software has been around since the 1950s, the launching of the GitHub platform in 2008 provided a collectable source of metadata that could be harvested and analyzed. Repositories with machine-detectable licenses that are approved by the OSI were identified, and public, active (non-archived) repositories that are not forks or mirrors of other repositories were collected using the *GHOST.jl* package (Santiago Calderon, 2020). The information collected for each repository includes creation date, description of the repository, primary language, user-selected topics, and a list of all developers that appear in the commit data for the identified repositories were compiled and a set of user data was collected for each developer, including GitHub account creation date, email address, location, bio, company, and website, if provided by the user.

A likely country of residence was then identified for each user based on available self-identified metadata using the *diverstidy* and *tidyorgs* packages (Kramer 2021a, 2021b). The initial implementation relied on only two fields (location and email domain), and returned a single

inferred country per contributor. This was subsequently expanded so that social accounts and the user bio can inform the country assignment, and that multiple countries can be identified for each user. An economic sector (academic, business, government, nonprofit) was also identified for developers based on the institution name in the user bio, affiliation, or email address.

Contributions to new repositories are an indicator of innovation. Both fractional counting and whole counting methods were employed for different analytical purposes. In fractional counts, each repository's credit is divided equally across the number of developers that contributed to it in the first year after its creation, and each developer's share is further divided across that developer's associated countries. If a developer is affiliated with two countries, each country receives half of that developer's share. This ensures that the total sum of all countries' contributions matches the total number of unique repositories. In whole counts, credit for each repository is counted one full time for each institution where at least one affiliate or employee contributed. Whole counting measures instances of collaboration and the contributions of particular institutions, but whole counts should not be aggregated because repositories codeveloped by multiple institutions would be double counted.

Limitations

The data collection aims to provide a census of all OSS projects on GitHub, but although GitHub is a large platform and is frequently used by developers for storing and distributing OSS, not all OSS projects are present on GitHub (Korkmaz et al. 2024). Furthermore, while there are many established OSS licenses, some valid OSS licenses may use nonstandard language that were not recognized by the *GHOST.jl* tool during data collection. Both of these issues may have resulted in an overall undercount of OSS projects.

A substantive caveat about the use of these data as indicators is that the GitHub platform does not maintain complete information about all developers. Location and sector analyses rely on location and affiliation data that GitHub users voluntarily supply, and the coverage of country and sector information is affected by the typical limitations of user-supplied data. Country could not be identified for 67% of GitHub user profiles; within the subset of U.S.-based GitHub users, sector could not be identified for 68% of profiles (<https://github.com/westat-oss>). As a result, data tables' geographic location and institutional affiliations are based only on the repositories with usable information. Data on the location of contributors is missing for more than half of the repositories (64%). Location information was available for between 32% and 52% of the repositories in each year from 2009 to 2024, with location information available for 36% of repositories in 2024 (Table STRN-33). In their Innovation Graph, which uses Internet Protocol data, GitHub research finds a similar distribution of GitHub repositories to those presented in the Indicators 2026 report Translation to Impact: U.S. and Global Science, Technology, and Innovation Output, with a large share of developers from the United States and India (<https://github.com/github/innovationgraph/blob/main/docs/datasheet.md>). Data on the institutional affiliation of contributors is missing for 84% of the repositories. This incomplete coverage may affect the reliability of location and sector analyses, which should be regarded as experimental at this time.

3. Supplemental Tables

In this section, Westat describes the supplemental tables associated with the OSS indicators presented in the *Indicators 2026* report “Translation to Impact: U.S. and Global Science, Technology, and Innovation Output” (NSB 2026a). The technical details regarding the production of these tables are provided in Section 4.

Table STRN-2: Contributions to New GitHub repositories by selected entities: 2009–24

Table STRN-2 shows federal agency contributions to OSS repositories over time and by agency. A repository is counted once for each federal agency in the first year that a member of the agency contributed to it. Whole counting is used for this table so that each agency’s contribution to a repository gets one count. The “Federal Total” reflects the total number of agency–repository combinations per year, counting each repository once per contributing agency. The “All Other Federal” row includes contributions from federal organizations not listed individually. Counts for specific agencies represent repositories with at least one contributor affiliated with that agency. The Department of Energy has consistently high contributions, and the steady rise in the federal total—from 14 repositories in 2009 to over 3,000 in 2024—illustrates growing government engagement in open-source development.

An earlier version of this table included source data from both GitHub and from a currently defunct federal website code.gov. Code.gov provided information about federal agency contributions to open-source software that included repositories located in other source code hosting platforms, such as Bitbucket and GitLab, in addition to GitHub. In 2025, code.gov was deactivated and the data ceased to be updated. As a result, the data in this table were recalculated in the Fall of 2025 to only include GitHub data.

Table STRN-33: New GitHub repositories contributed to, by selected region, country, or economy: 2009–24

Table STRN-33 shows a location-based tabulation of new GitHub OSS repository contributions in between 2009 and 2024 using fractional counting. Contributions to new repositories are an indicator of innovation. Each repository’s credit is divided equally across the number of its contributors in the first year after its creation, and each contributor’s share is further divided across their associated countries. If a contributor is affiliated with two countries, each country receives half of that contributor’s share. This ensures that the total sum of all countries’ contributions matches the total number of unique repositories.

The data in this table are used in Figures TRN-A and TRN-B. Figure TRN-A shows the growth in the number of new OSS repositories on GitHub since 2009, one year after GitHub began. The figure shows the growth from US contributors separately from the rest of the world. Figure TRN-B shows the shares of new repositories contributed to for the 10 locations with the highest number of contributions for 2024.

Table STRN-34: Total GitHub repositories contributed to by selected entities: 2009–24

Table STRN-34 shows the top 10 business and top 10 academic institutions contributing to GitHub repositories across the years 2009–2024, with their location in parentheses. The table uses a whole-counting method; each repository is counted once for every institution that contributed to it, determined by the organization affiliations of the users. The business total and academic total rows represent the total number of repository-organization combinations involving business and academic institutions, respectively, including those from the top 10. Each institution is counted once per repository where it had at least one contributor, based on the earliest year of involvement. The all other business and all other academic rows capture contributions from institutions not listed in the respective top 10 lists.

Table STRN-35: Open-source software collaboration networks in GitHub, by region, country, or economy: 2024

Collaboration between developers in different locations is one indicator of the existence of partnership relationships across countries. Table STRN-35 shows collaboration pairs with more than 1,000 co-developed repositories between the two countries in one year. This table is calculated with whole counting, so each collaboration pair counts as one. Since a repository can have several country pairs collaborating, the total possible number of pairs exceeds the number of repositories.

Figure TRN-C uses these data for a collaboration network visualization that follows a similar methodology to the collaboration network that visualized coauthored semiconductor articles in the Discovery report (NSB 2025). For illustrative purposes, Figure TRN-C shows edges with greater than 1,000 co-developed repositories.

Table STRN-36: Open-source software repositories contributed to in GitHub, by country of contributor: 2024

Table STRN-36 shows a country-based tabulation of all GitHub OSS repository contributions in between 2024 using fractional counting. Each repository’s credit is divided equally across the number of its contributors, and each contributor’s share is further divided across their associated countries. If a contributor is affiliated with two countries, each country receives half of that contributor’s share. This ensures that the total sum of all countries' contributions matches the total number of unique repositories. The data in this table are used to represent node sizes in the collaboration network visualization (Figure TRN-C).

4. Methodology for Construction of Figures and Supplemental Tables

1.1 Data Collection

In this section, Westat describes the GitHub data pipeline, including details about data collection, processing, sectoring and development of data tables and figures shown in Table 1. Westat leveraged GHOST.jl, an open-sourced Julia package that uses the GitHub v4 GraphQL API, to collect data on GitHub repositories and developers. Figure 1 illustrates the data pipeline for GitHub, which includes details about data collection, processing, sectoring and development of data tables.

4.1.1 LICENSES. The first step of data collection using GHOST.jl is to identify repositories with machine-detectable licenses that are approved by the Open Source Initiative (OSI). The list of OSI-approved licenses are published as SPDX License List data by the SPDX Working Group. The script `licenses.jl` is used to generate the licenses table.

4.1.2 QUERIES. After the licenses table is created, the next step is to run `queries.jl` to construct queries for repositories that are using these licenses. Westat limits the collection to public non-archived (active) repositories that are not forks or mirrors of other repositories. GitHub only allows to query up to 1,000 results per search connection result, we develop queries to include 1,000 repositories at each API call. If a query returns over 1,000 results, only the first 1,000 are accessible. To collect every repository of interest, Westat shrinks intervals until the result count is 1,000 or fewer. The queries table is used to store the queries and track their status.

4.1.3 REPOS. The next step after constructing a table of queries is to collect the data for each repository using the script `repos.jl`. The repositories are represented by *slugs* which are URL-friendly, lowercase, hyphen-separated strings that uniquely identify each repository. The information collected includes creation date, description of the repository, primary language, user-selected topics, and metrics such as number of forks, dependencies stargazers, watchers and commits.

4.1.4 COMMITS. For each repository, Westat queries the commit data for years between 2009 and 2024. Commits are approved code changes (additions and deletions of lines of code) by developers in GitHub repositories. The commit data for a GitHub repository is dependent on the base branch (identified by unique branch IDs). The script `commits.jl` selects repositories from the previous step and collects commit data in batches (16/32/64 commits at a time) depending on the size and number of commits for each repository.

4.1.5 USERS. Finally, the script `users.jl` compiles a list of all developers that appear in the commit data and collects user data for each author including account creation date, email address, location, bio, company, website, if provided by the user.

1.2 Assignment of Countries, Institutions and Sectors to Users

In this section, Westat describes how country, institution and sector information is assigned to individual GitHub users. Westat built on two open-source packages written in R:

- **Diverstidy.R** — used for **country assignment**
- **Tidyorgs.R** — used for institution and **sector assignment**

Sections 4.2.1 and 4.2.2 briefly describe each package and the enhancements made to improve data accuracy, correct classification errors, and support comprehensive assignment of countries and institutional sectors. Section 4.2.3 describes how outputs from both packages were merged to develop the final user table.

4.2.1 diverstidy.R: This package assigns a country to each user based on available metadata provided by the developers. The initial implementation relied on only two fields (location and email domain) and returned a single inferred country per contributor. Westat modified the script, and the inputs used by the updated version and the output generated are provided in Table 2 below. Moreover, each output is now source-specific (the script prints the source of information for specific country assignments), allowing greater transparency, cross-validation, and error resolution.

Table 2. Inputs and outputs for diverstidy.R	
Input	Description
Location	Self-reported location in the user profile (e.g., city, country)
Email	Email address linked to the GitHub account
Social accounts	Username for social accounts such as X (formerly Twitter) and LinkedIn
Bio	Bio section in the user profile
Output	Description
country_location	country derived from location string
country_email	country derived from email domain
country_socialaccounts	country derived from linked social accounts
country_bio	country derived from biography text on GitHub

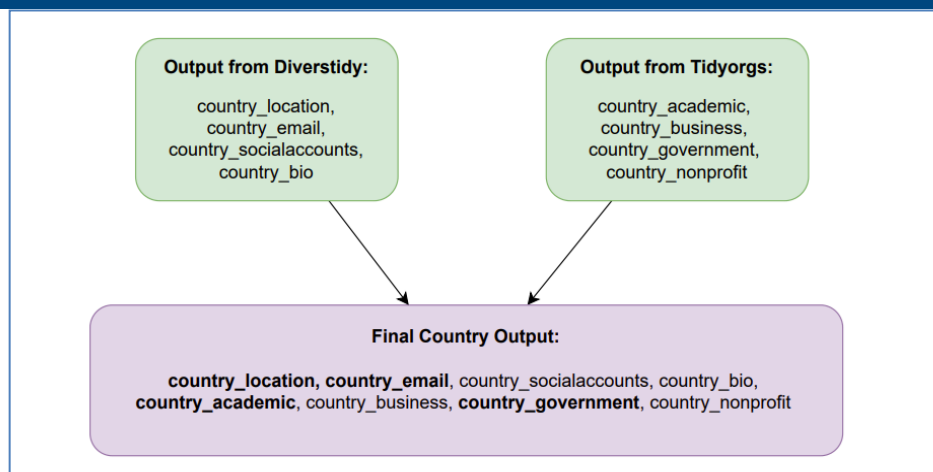
4.2.2 tidyorgs.R: This package assigns an economic sector (academic, business, government, nonprofit) to developers based on the identified institution name. Westat incorporated external institutional data (Sector_Mapping_Institutions_Patent_Assignees.csv) received from NCSES for broader coverage of known entities, especially in academic, business, and government sectors. The previous version of the code used self-reported company field (organization or affiliation field on GitHub), and email address (domain used to infer affiliation on GitHub) provided by the developers, and generated binary variables for each sector. However, the source of the assignment was not printed as an output. Westat updated the code to generate the outputs listed in Table 3 below.

Table 3. Input and outputs for tidyorgs.R	
Input	Description
Company	Self-reported organization in the user profile
Email	Email address linked to the GitHub account
Output	Description
is_academic, is_business, is_government, is_nonprofit	Binary variables for each sector
organization_company_academic, organization_email_academic, organization_company_business, organization_email_business, organization_company_government, organization_email_government, organization_company_government, organization_email_government	Source-specific sector information
country_academic, country_business, country_government, country_nonprofit	Country identified from organizations

4.2.3 Developing the User Table. The final step for developing the user table is to merge the outputs generated by both packages. The final user table includes country and sector assignments and the sources of information used for these assignments. Figure 1 illustrates this process for country assignment. This merge allows us to identify and resolve conflicts for country and sector assignments. Westat also conducted exploratory data analysis to assess data completeness, to evaluate source agreement, and to validate the impact of the updates on attribution coverage and accuracy.

These enhancements substantially improved the quality and transparency of contributor classification, supporting more reliable analysis of open-source participation by geography and institutional sector.

Figure 1. Country Assignment



1.3 Data Cleaning and Processing

This section outlines the data cleaning, processing, and integration steps taken to prepare GitHub data for the generation of the tables and figures. The goal is to prepare a cleaned dataset of users and their repository-level activity, with organization names, country labels, and sector assignments. Table 4 below lists the steps taken for user and commit data tables.

Table 4. Data cleaning, processing, and integration steps	
User Data	Cleaning and Formatting Steps
	The raw GitHub user dataset with country and sector fields was first subset to relevant columns and then cleaned using DuckDB
	Four fields were used for country assignment: country_location, country_email, country_government, and country_academic. These were merged, deduplicated, and collapsed into a unified field country_cleaned.
	Organization fields (from both email domains and free-text inputs) were merged into a combined field organization_cleaned, with common suffixes (e.g., “Inc.”) removed.
	The resulting cleaned dataset was written to the file: <i>user_data_country_sectors_cleaned.parquet</i>
Commit Data (2009–2024)	Deduplication and Aggregation
	Raw GitHub commit files (2009–2024) were processed year by year
	For each year, the code: • Read all commits • Unnested the list of author IDs • Extracted the commit year from the timestamp • Returned a deduplicated list of (branch_id, author_id, commit_year) combinations
	These yearly files were saved as: <i>unique_commits_<YYYY>.parquet</i> where YYYY goes from 2009 to 2024
	A final aggregate file was created by: • Combining all yearly commits • Grouping by author_id and branch_id • Calculating min_commit_year to capture the earliest year of contribution per user-repository pair
	This yielded the output file: <i>unique_commits_2009_2024.parquet</i>

1.4 Production of Final Data Tables and Figures

This section documents the creation of the data tables and figures developed for the OSS indicators in the *Indicators 2026* report. Each table is derived from processed data collected from GitHub, which includes standardized user- and commit-level information with country, institution and sector classifications. Westat describes the logic used to generate each table, based on filtered subsets, aggregations, and joins from the cleaned datasets. Each table is presented in its own subsection, with clear documentation of inputs, transformation logic, and intended use.

Table STRN-2: Contributions to New GitHub repositories by selected entities: 2009–24

This table reports the number of new GitHub repositories contributed to by selected U.S. federal government agencies from 2009 to 2024, using a whole-counting method. A repository is counted once for each federal agency that contributed to it, based on the earliest year of

agency involvement. The federal total reflects the total number of agency-repository combinations per year, counting each repository once per contributing agency. All other federal row includes contributions from federal organizations not listed individually. Counts for specific agencies represent repositories with at least one contributor affiliated with that agency. A repository may be counted for multiple agencies if multiple agencies contributed during its first year of federal activity. Table 5 below Westat describes the steps taken to develop the table.

Table 5. Data processing steps for Table STRN-2	
Input Datasets	
Dataset 1	user_data_country_sectors_cleaned.parquet: Cleaned and merged user-level metadata from GitHub, including sector and organization assignments.
Dataset 2	unique_commits_2009_2024.parquet: Aggregated commit records capturing the first year of contribution (min_commit_year) for each user-repository.
Data Processing Steps	
1. User Data Expansion and Filtering	- The organization_cleaned, country_cleaned, and sector fields are each split and unnested to allow one value per row. - The dataset is then filtered to retain only contributors from the United States with an assigned government sector.
2. Joining Users with Aggregated Commits	- Commits are joined to filtered users by user ID. - For each (branch, organization) pair, the earliest year of contribution is determined using the min_commit_year value. - The resulting dataset identifies the first commit year of each organization in each repository. This enables analysis of how many repositories each agency engaged with in each year of the time series.
3. Agency-Level Aggregation	- For each agency in a predefined list of 20 major U.S. federal entities, the number of newly contributed repositories is counted per year. - Organizations not on the list are grouped under the category “All Other Federal.”
4. Federal Total Aggregation	- A separate count is computed to reflect the total number of new repositories with any federal participation in a given year, regardless of agency. - This row appears first in the final output as “Federal Total.”
5. Pivoting to Wide Format	- All three aggregations (agency-specific, federal total, and other federal) are reshaped so that each row corresponds to an agency and each column corresponds to a year (2009–2024). - Missing combinations are filled with 0s.
6. Final Row and Column Ordering	- The agency rows are arranged in a fixed order, with “Federal Total” listed first and “All Other Federal” listed last. - Year columns are sorted chronologically from 2009 to 2023.
Final Table Format	
	- A column Institution with agency names, “Federal Total,” and “All Other Federal” - One column for each year from 2009 to 2024, containing integer counts of new repositories contributed to by the listed entity

Table STRN-33: New GitHub repositories contributed to, by selected region, country, or economy: 2009–24

This table provides a time series summary of global participation in new GitHub repositories from 2009 to 2024, highlighting the ten most active contributing countries. Contributions are measured using fractional repository credit, allocated based on contributor presence in a repository’s first year of activity. Additional summary rows are included to capture total participation, contributions from all other countries, and missing data.

The data in the table STRN-33 is used to generate Figure TRN-A and TRN-B. Figure TRN-A shows the growth in the number of new OSS repositories on GitHub between 2009 and 2024. The figure shows US contributions separately from the rest of the world. Note that Figure TRN-A includes repositories that can be attributed to at least one country, i.e., the sum of contributions from the US and the rest of the world in a year corresponds to the Total (Any Country) row in STRN-33. Figure TRN-B shows the shares of new repositories contributed to for the 10 locations with the highest number of contributions for 2024. The fractions are calculated by dividing each country’s repository contributions in 2024 by the corresponding Total (Any Country) row in STRN-33.

Table 6. Data processing steps for Table STRN-33	
Input Datasets	
Dataset 1	user_data_country_sectors_cleaned.parquet: Cleaned GitHub user-level metadata with country information.
Dataset 2	unique_commits_2009_2024.parquet: Aggregated commit records with the earliest contribution year per user–repository pair.
Data Processing Steps	
1. User Data Expansion and Filtering	- The country_cleaned field is split and unnested so that each contributor-country pair is on a separate row. - Missing values are labeled “Missing”.
2. Joining Users with Aggregated Commits	- Commits are joined to expanded user-country data by contributor ID (author_id). - Only repositories with at least one user-country match are retained. - The earliest contribution year for each repository is identified using min_commit_year.
3. Fractional Credit Assignment	- Each repository’s credit is distributed evenly across all users active in its first year. - For contributors associated with multiple countries, this credit is further divided equally among countries.

Input Datasets	
4. Aggregation by Country and Year	• Generate Total Repos Row: Sum of fractional credit across all contributors, regardless of country. • Generate Total (Any Country) Row: Sum of credit from contributors with known country affiliations. • Select Top 10 Countries: Determined by aggregating each country's total fractional credit across all years. Annual contributions are then tabulated for these top 10 countries. • All Other Countries: Contributions from countries outside the top 10 (excluding “Missing”). • Total (Missing Country): Credit from contributors without a known country.
5. Pivoting and Formatting	• All summaries are pivoted so that each row corresponds to a country or total category, and columns represent years 2009 through 2024. • Values are rounded to whole numbers for presentation.
6. Final Row and Column Ordering	• The agency rows are arranged in a fixed order, with “Federal Total” listed first and “All Other Federal” listed last. • Year columns are sorted chronologically from 2009 to 2023.
Final Table Format	
	• A column “Country” with the entities listed in the order displayed above • One column for each year from 2009 to 2024, containing fractional counts of new repositories contributed to by the listed entity, and is restricted to contributors active in a repository’s first year only • The total row (Total Repos) represents the true number of distinct repositories, summing to the same value across all years.

Table STRN-34: Total GitHub repositories contributed to by selected entities: 2009–24

This supplemental table presents the top institutional contributors to OSS among business and academic entities worldwide. Using the same whole-counting method as in STRN-2, the table counts the total number of unique GitHub repositories each organization has contributed to across the 2009–2024 period. The top 10 business organizations and top 10 academic institutions are displayed, along with category totals and catch-all “all other” rows.

Table 7. Data processing steps for Table STRN-34	
Input Datasets	
Dataset 1	user_data_country_sectors_cleaned.parquet: Cleaned GitHub user-level data with organization, country, and sector fields.
Dataset 2	unique_commits_2009_2024.parquet: Aggregated commit records capturing the first year of contribution (min_commit_year) for each user-repository.
Data Processing Steps	

Input Datasets	
1. User Data Expansion and Filtering	- The organization_cleaned, country_cleaned, and sector fields are each split and unnested to allow one value per row. - The dataset is then filtered to retain only contributors whose assigned sector is "business" or "academic".
2. Joining Users with Aggregated Commits	- Commits are joined to filtered users by user ID. - For each (branch, organization) pair, the earliest year of contribution is determined using the min_commit_year value (aggregation is cumulative across all years).
3. Organization - Level Aggregation	- The total number of distinct repositories (branches) is counted for each unique organization within each sector. - This forms the basis for ranking top contributors.
4. Subsetting and Summarization	- Select Top 10 Businesses: - The 10 business organizations with the most repositories are selected. - The remaining business entities are grouped into a summary row labeled "All Other Business." - A category-level total labeled "Business Total" is also computed. - Select Top 10 Academic Institutions: - A similar process is followed for academic contributors. - A summary row "All Other Academic" and a category-level "Academic Total" are included.
Final Table Format	
	- A column "Institution" with "Business Total", Top 10 Businesses, "All Other Businesses", "Academic Total", Top 10 Universities, and "All Other Academic." - A column "Total Repositories" showing the counts of new repositories contributed to by the listed entity over the period 2009-2024.

Table STRN-35: Open-source software collaboration networks in GitHub, by region, country, or economy: 2024

This table presents a pairwise count of international collaborations in OSS development based on co-authorship of GitHub repositories, using a whole-counting method. A collaboration is defined as a repository (branch) with contributors from at least two different countries. The output lists country pairs (edges) and the number of repositories they co-developed for country-pairs that have at least 1,000 co-developed repositories. Note that this table uses the commit data for 2024 and includes all repositories committed to by users in 2024 (not only new repositories as in STRN-33). This table enables the construction of a network graph representing international ties in open-source development, based on shared repository authorship.

Table 8. Data processing steps for Table STRN-35	
Input Datasets	
Dataset 1	user_data_country_sectors_cleaned.parquet: Cleaned GitHub user-level metadata with country information.
Dataset 2	unique_commits_2024.parquet: Aggregated commit records linking users to repositories in 2024.
Data Processing Steps	
1. User Data Expansion and Filtering	- The country_cleaned field is split on commas and unnested to assign one country per row per user. - Missing values are labeled as “Missing Country.”
2. Joining Users with Aggregated Commits	The 2024 commit data (filtered to include only records from 2024) is joined with the user-country pairs by user ID to generate a table of (branch, country, author_id) combinations.
3. Generating Collaboration Pairs (Edges)	- Identify unique combinations of repository (branch), contributor, and country. - For each branch, identify the full list of contributing countries and filter out repositories with only one contributor (no collaboration). - For each branch, identify all unique country pairs (all pairwise combinations). For example, for a branch with contributors from Germany, Japan, and Brazil, the code will generate three pairs: (Brazil, Germany), (Brazil, Japan), (Germany, Japan). - Filter out repositories with contributors from a single country.
4. Computing Repository Counts per Collaboration Pair (Edge Weights)	- The country pairs are grouped by counting their occurrences to produce the total number of distinct co-developed repositories (branches) per pair. - Country pairs containing "Missing Country" are excluded. - Final edge-list is sorted in descending order by the number of co-developed repositories.
5. Removing Outliers	Westat identified four users with contributions exceeding 10,000 repositories in 2024, these users were treated as outliers and excluded from subsequent tables and the network figure.
Final Table Format	
	Columns “Country 1”, “Country 2”, and “Co-developed repositories”

Table STRN-36: Open-source software repositories contributed to in GitHub, by country of contributor: 2024

This table provides a fractional count of repositories attributed to each country based on contributor activity. Rather than assigning full credit to each contributor’s country, the function distributes fractional repository credit across countries proportionally, accounting for both

multi-user and multi-country participation. This ensures that the total number of repositories is preserved when aggregated.

Table 9. Data processing steps for Table STRN-36	
Input Datasets	
Dataset 1	user_data_country_sectors_cleaned.parquet: Cleaned GitHub user-level data with country assignments, including users with multiple countries
Dataset 2	unique_commits_2024.parquet: Aggregated commit records linking users to repositories in 2024.
Data Processing Steps	
1. User Data Expansion and Filtering	- The country_cleaned field is split on commas and unnested to assign one country per row per user. - Missing values are labeled as “Missing Country.”
2. Joining Users with Aggregated Commits	- The 2024 commit data is joined with the user dataset using the author_id. - Branches (repositories) that include at least one contributor with a known country are retained for further analysis.
3. Fractional Credit Assignment	- For each repository (branch), credit is distributed equally across its unique contributors. For example, a repository with 4 contributors gives each user a credit of 0.25. - If a contributor is associated with multiple countries, their user-level credit is further split equally across those countries. For example, a user from both Canada and Germany who receives 0.25 credit on a repository will contribute 0.125 credit to each country.
4. Country-Level Aggregation	- For each country, fractional credits are summed across all repositories. - The result is a single fractional repository total for each country that reflects multi-national contributions.
5. Summarization	- Calculate Total Repos: The overall sum of fractional credit across all countries (should match the number of distinct repositories). - Calculate Total (Missing Country): Total repository credit associated with contributors whose country is unknown. - Calculate Total (Any Country): Total repository credit associated with known countries only.
Final Table Format	
	- Columns “Country”, “Total repositories”, and “Co-developed repositories.” - The first three rows of the table contain summary totals as described above. The remainder of the table lists countries in descending order of total fractional repository credit.

Figure TRN-C: Open-source software collaboration networks in GitHub, by region, country, or economy: 2024

Figure TRN-C is a network visualization of collaborations between countries that uses STRN-35 to illustrate the edge weights (measured by the number of co-developed repositories) and STRN-35 to capture the size of the nodes (total number of repositories for each country). The network visualization was generated similarly to the collaboration figure used in the Discovery report for bibliometric collaborations (NSB 2025). For illustrative purposes, the network shows edges greater than 1,000. The final product includes three distinct network graphs, each tailored to emphasize a specific aspect of international OSS collaboration: (1) overall collaboration structure, (2) repository leadership by country, and (3) the strongest bilateral relationships.

Table 10. Generating Figure TRN-C	
Input Datasets	
Dataset 1	Table STRN-35 (edge list and weights): pairs of countries and the number of co-developed repositories in 2024.
Dataset 2	Table STRN-36 (node size): fractional repository counts per country in 2024.
Dataset 3	countries_data.rda from the diverstidy repository: used for continent mapping and name standardization (available at https://github.com/westat-oss/diverstidy/blob/main/data/countries_data.rda)
Packages	
	igraph: graph structure creation and metrics (e.g., degree)
	ggraph, ggplot2: graph layout and plotting
Data Processing Steps	
1. Country Standardization	- Country names were standardized across both node and edge datasets to ensure consistency. - Manual standardization where necessary (e.g., mapping "Russian Federation" and "Germany" to consistent identifiers) to be consistent with other Indicators tables.
2. Continent Assignment	- Countries were mapped to continents using the diverstidy.R reference data. - Missing continent values were manually filled in for a small number of countries
Graph Construction	
	- An undirected graph was created using <code>igraph::graph_from_data_frame()</code>, with edges representing co-developed repositories and nodes representing countries. - Node attributes: node size (STRN-36), node degree (number of unique collaborations per country), and continent. - Edge Filter: Included only country pairs with more than 1,000 co-developed repositories. - Layout coordinates were manually adjusted to improve interpretability.

References

- GitHub. 2026a. Let's Build from Here. Available at <https://github.com/about>. Accessed 10 April 2026.
- GitHub. 2026b. Git Commit. Available at <https://github.com/git-guides/git-commit>. Accessed 10 April 2026.
- Korkmaz G, Santiago Calderón JB, Kramer BL, Guci L, Robbins CA. 2024. From GitHub to GDP: A Framework for Measuring Open Source Software Innovation. *Research Policy* 53(3):104954. Available at <https://doi.org/10.1016/j.respol.2024.104954>. Accessed 10 April 2026.
- Kramer BL. 2021a. *diverstidy*: A Tidy Package for Detection and Standardization of Geographic, Population, and Diversity-Related Terminology in Unstructured Text Data. Available at <https://www.brandonleekramer.com/packages/>. Accessed 23 March 2026.
- Kramer BL. 2021b. *tidyorgs*: A Tidy Package That Standardizes Text Data for Organizational Analysis. Available at <https://www.brandonleekramer.com/packages/>. Accessed 23 March 2026.
- National Science Board (NSB), National Science Foundation. 2022. Invention, Knowledge Transfer, and Innovation. *Science and Engineering Indicators 2022*. NSB-2022-4. Alexandria, VA. Available at <https://ncses.nsf.gov/pubs/nsb20224/>. Accessed 4 May 2026.
- National Science Board (NSB), National Science Foundation. 2025. Discovery: R&D Activity and Research Publications. *Science and Engineering Indicators 2026*. NSB-2025-7. Alexandria, VA. Available at <https://ncses.nsf.gov/pubs/nsb20257/>.
- National Science Board (NSB), National Science Foundation. 2026a. Translation to Impact: U.S. and Global Science, Technology, and Innovation Output. *Science and Engineering Indicators 2026*. NSB-2026-2. Alexandria, VA. Available at <https://ncses.nsf.gov/pubs/nsb20262>. Accessed 4 May 2026.
- National Science Board (NSB), National Science Foundation. 2026b. Translation to Impact: U.S. and Global Science, Technology, and Innovation Output. *Science and Engineering Indicators 2026*. NSB-2026-2. Alexandria, VA. Available at <https://ncses.nsf.gov/pubs/nsb20262/supplemental-tables>. Accessed 4 May 2026.
- Santiago Calderón JB. 2020. GHOST.jl. Available at <https://github.com/team-oss/GHOST.jl>. Accessed 4 May 2026.